

```
!pip install pandas
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from pandas)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil)
```

```
import pandas as pd
import numpy as np
```

```
print(pd.__version__)
```

```
2.2.2
```

```
s=pd.Series([1,2,3,4])
print(s)
```

```
0    1
1    2
2    3
3    4
dtype: int64
```

```
#from List
s2=pd.Series([1,2,3,4],index=['a','b','c','d'])
print(s2)
```

```
a    1
b    2
c    3
d    4
dtype: int64
```

```
student={
    "name":"vamshi",
    "branch":"aiml"
}
se=pd.Series(student)
print(se)
```

```
name      vamshi
branch    aiml
dtype: object
```

```
n=pd.Series([10,20,30,40])
```

```
n+2
```

	θ
0	12
1	22
2	32
3	42

dtype: int64

n-2

	θ
0	8
1	18
2	28
3	38

dtype: int64

n*2

	θ
0	20
1	40
2	60
3	80

dtype: int64

n%2

	θ
0	0
1	0
2	0
3	0

dtype: int64

```
n=pd.Series([10,20,30,40])
```

```
print(n.max)
print(n.min)
print(n.mean)
print(n.sum)
```

```
print(n.median)
print(n.std)
```

```
<bound method Series.max of 0      10
1      20
2      30
3      40
dtype: int64>
<bound method Series.min of 0      10
1      20
2      30
3      40
dtype: int64>
<bound method Series.mean of 0      10
1      20
2      30
3      40
dtype: int64>
<bound method Series.sum of 0      10
1      20
2      30
3      40
dtype: int64>
<bound method Series.median of 0      10
1      20
2      30
3      40
dtype: int64>
<bound method Series.std of 0      10
1      20
2      30
3      40
dtype: int64>
```

```
n=pd.Series([10,20,30,40])
```

```
s
```

```
#vectorized string
names=pd.Series(['vishnu','teja','manoj'])
```

```
names.str.lower()
names.str.upper()
names.str.len()
names.str.startswith('h')
```

```

      0
-----
0  False
1  False
2  False

dtype: bool
```

```
std=[
    {"name":"vishnu","branch":"aiml","age":20},
    {"name":"teja","branch":"aiml","age":21},
    {"name":"manoj","branch":"aiml","age":21}
]
```

```
s=pd.DataFrame(std)
print(s)
```

	name	branch	age
0	vishnu	aiml	20
1	teja	aiml	21
2	manoj	aiml	21

```
#From numpy array
arr=np.random.rand(3,4)
c=pd.DataFrame(arr,columns=['A','B','C','D'],index=(1,2,3))
print(c)
```

	A	B	C	D
1	0.406162	0.264374	0.451636	0.926568
2	0.682741	0.025500	0.204949	0.637638
3	0.067217	0.577487	0.955189	0.094927

```
#exploring data frame
data={
    'name':['vishnu','teja','manoj'],
    'age':[20,21,21],
    'score':[23,34,45],
    'grade':['A','B','C']
}
df=pd.DataFrame(data)
print(df)
```

	name	age	score	grade
0	vishnu	20	23	A
1	teja	21	34	B
2	manoj	21	45	C

```
df.head(2)
```

	name	age	score	grade
0	vishnu	20	23	A
1	teja	21	34	B

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.tail(2)
```

	name	age	score	grade
1	teja	21	34	B
2	manoj	21	45	C

```
df.nunique()
```

```
      0
name    3
age     2
score   3
grade   3

dtype: int64
```

```
df.isnull().sum()
```

```
      0
name    0
age     0
score   0
grade   0

dtype: int64
```

```
df.dtypes
```

```
      0
name  object
age   int64
score int64
grade object

dtype: object
```

```
df.index
```

```
RangeIndex(start=0, stop=3, step=1)
```

```
df["name"]
```

```
      name
0  vishnu
1    teja
2  manoj

dtype: object
```

```
df.name
```

```
      name
0  vishnu
1    teja
2  manoj
```

dtype: object

```
df[["name", "age"]]
```

```
      name  age
0  vishnu   20
1    teja   21
2  manoj   21
```

```
df.loc[0]
```

```
      0
name  vishnu
age    20
score   23
grade    A
```

dtype: object

```
df.loc[0:2]
```

```
      name  age  score  grade
0  vishnu   20     23     A
1    teja   21     34     B
2  manoj   21     45     C
```

```
df.iloc[0]
```

```
      0
name  vishnu
age    20
score   23
grade    A
```

dtype: object

```
df.loc[0:2, ["name", "age"]]
```

	name	age
0	vishnu	20
1	teja	21
2	manoj	21

```
df.iloc[0:2,0:2]
```

	name	age
0	vishnu	20
1	teja	21

```
df.iloc[0:2]
```

	name	age	score	grade
0	vishnu	20	23	A
1	teja	21	34	B

```
df.iloc[-1]
```

	2
name	manoj
age	21
score	45
grade	C

dtype: object

```
df.loc[0, 'name']
```

'vishnu'

```
df.iloc[0,0]
```

'vishnu'